

CST444	SOFT COMPUTING	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
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COURSE OBJECTIVE

To introduce the concepts in Soft Computing such as Artificial Neural Networks, Fuzzy logic based systems, genetic algorithm-based systems and their hybrids.

SYLLABUS

Module – 1 (Introduction to Soft Computing & Artificial Neural Network)

Introduction to Soft Computing. Difference between Hard Computing & Soft Computing. Applications of Soft Computing. Artificial Neurons Vs Biological Neurons. Basic models of artificial neural networks – Connections, Learning, Activation Functions. McCulloch and Pitts Neuron. Hebb network.

Module – 2 (Supervised Learning Network)

Perceptron Networks– Learning rule, Training and testing algorithm. Adaptive Linear Neuron– Architecture, Training and testing algorithm. Back propagation Network – Architecture, Training and testing algorithm.

Module - 3 (Fuzzy Logic & Defuzzification)

Fuzzy sets – properties, operations on fuzzy set. Fuzzy membership functions, Methods of membership value assignments – intuition, inference, Rank Ordering. Fuzzy relations– operations on fuzzy relation. Fuzzy Propositions. Fuzzy implications. Defuzzification– Lamda cuts, Defuzzification methods.

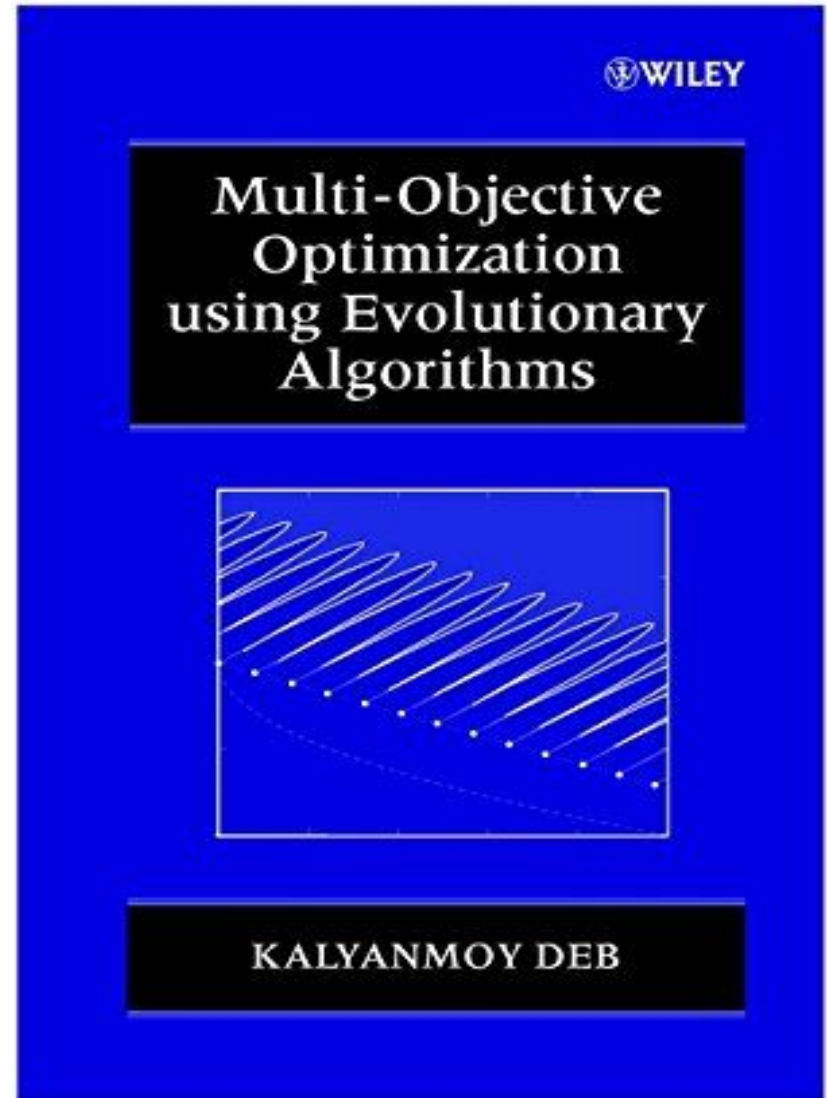
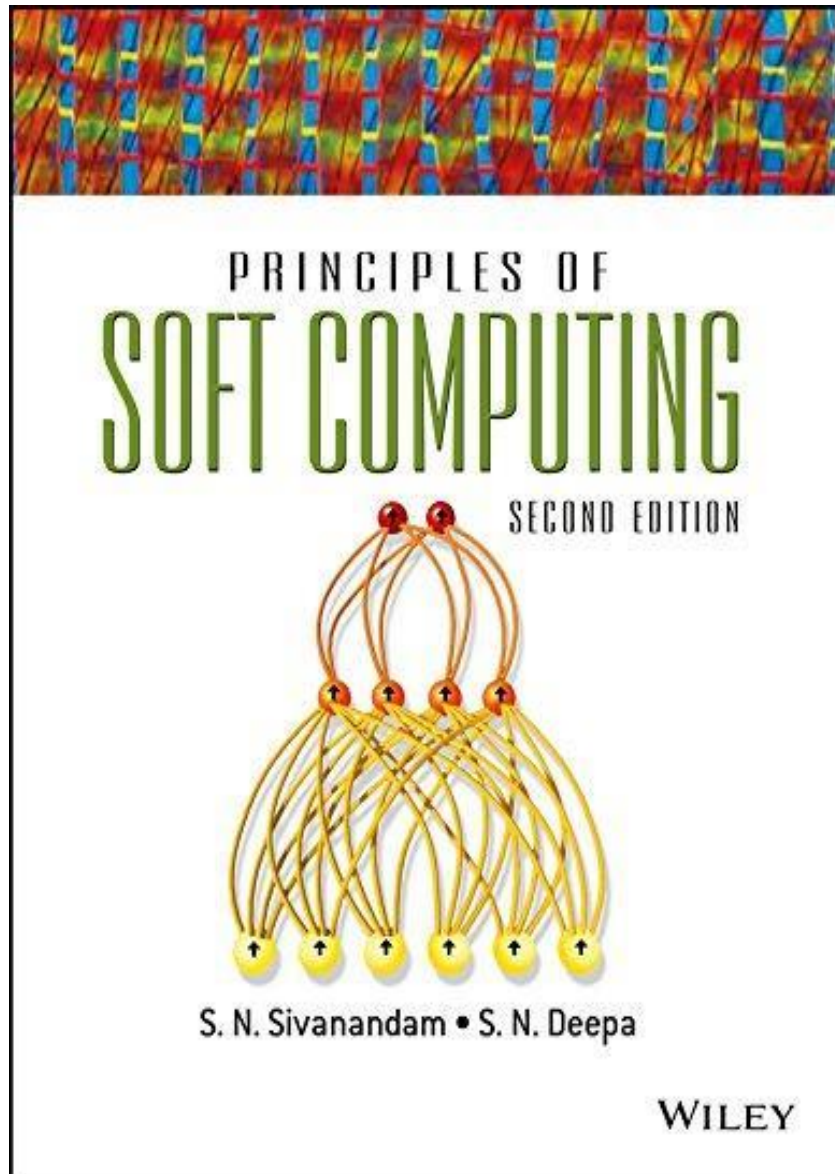
Module - 4 (Fuzzy Inference System & Genetic Algorithm)

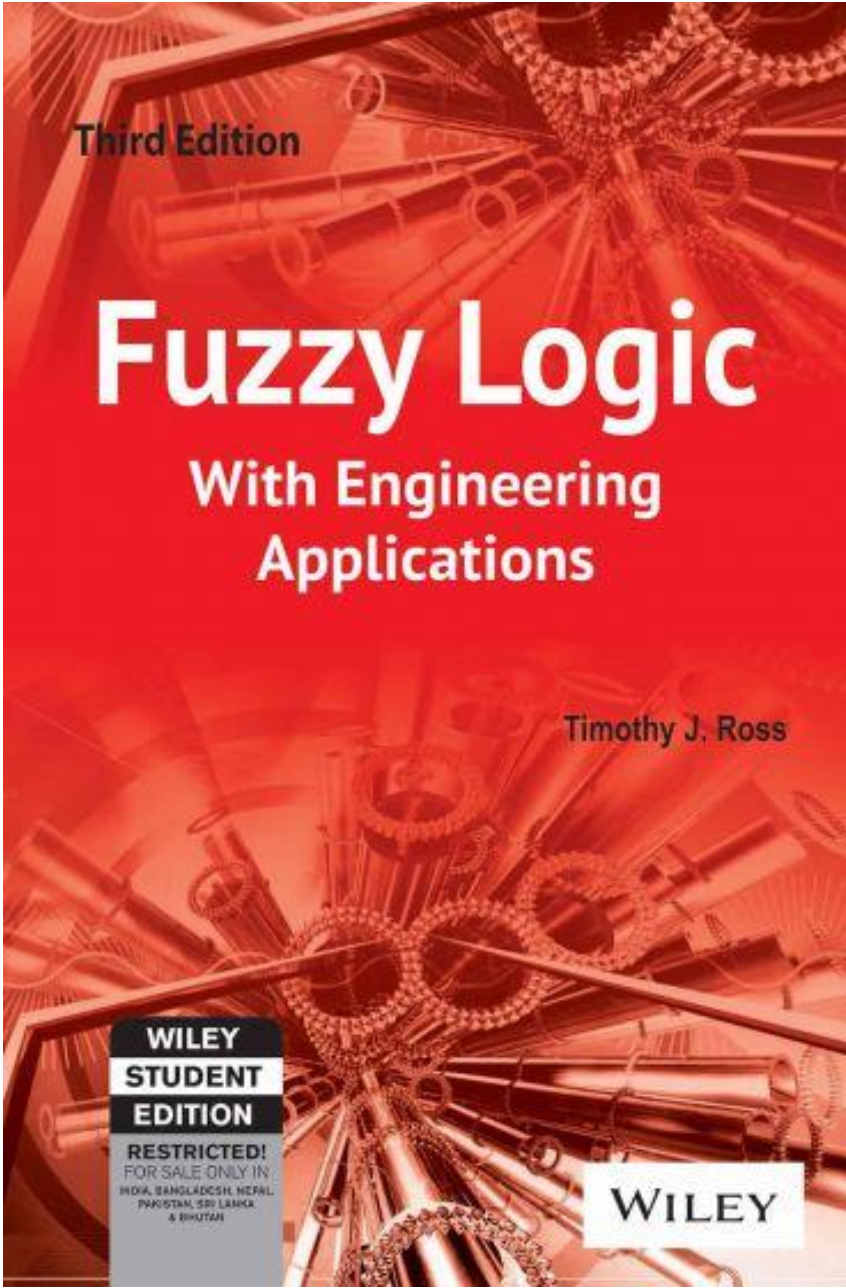
Fuzzy Inference Systems - Mamdani and Sugeno types. Fuzzy Logic Controller. Concepts of genetic algorithm. Operators in genetic algorithm - coding, selection, cross over, mutation. Stopping condition for genetic algorithm.

Module - 5 (Multi Objective Optimization & Hybrid Systems)

Multi objective optimization problem. Principles of Multi- objective optimization, Dominance and pareto-optimality. Optimality conditions. Neuro-fuzzy hybrid systems. Genetic – neuro hybrid systems.

BOOKS REFERRED





Third Edition

Fuzzy Logic

With Engineering
Applications

Timothy J. Ross

WILEY
STUDENT
EDITION

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INDIA, BANGLADESH, NEPAL,
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COURSE OUTCOME

CO1	Describe soft computing techniques and the basic models of Artificial Neural Network (Cognitive Knowledge Level: Understand)
CO2	Solve practical problems using neural networks (Cognitive Knowledge Level: Apply)
CO3	Illustrate the operations, model and applications of fuzzy logic (Cognitive Knowledge Level: Apply)
CO4	Illustrate the concepts of Genetic Algorithm (Cognitive Knowledge Level: Apply)
CO5	Describe the concepts of multi-objective optimization models and the need for using hybrid soft computing approaches(Cognitive Knowledge Level: Understand)

SOFT COMPUTING (SC)

Soft Computing (SC): The symbiotic use of many emerging problem-solving disciplines.

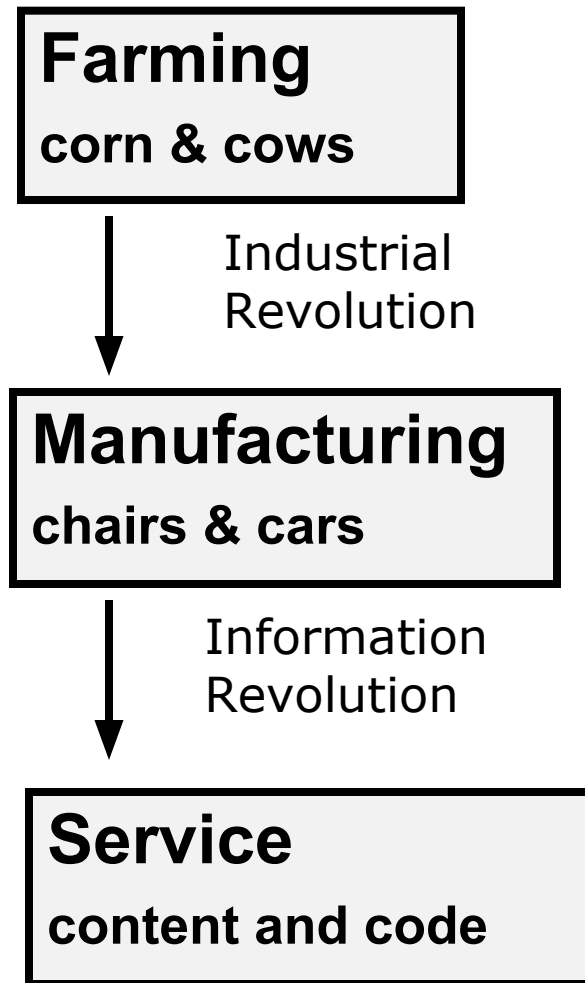
According to Prof. Zadeh:

"...in contrast to traditional hard computing, soft computing exploits the tolerance for imprecision, uncertainty, and partial truth to achieve tractability, robustness, low solution-cost, and better rapport with reality"

Soft Computing Main Components:

- Approximate Reasoning
- Search & Optimization
 - ✓ Neural Networks, Fuzzy Logic, Evolutionary Algorithms

Why is this important?



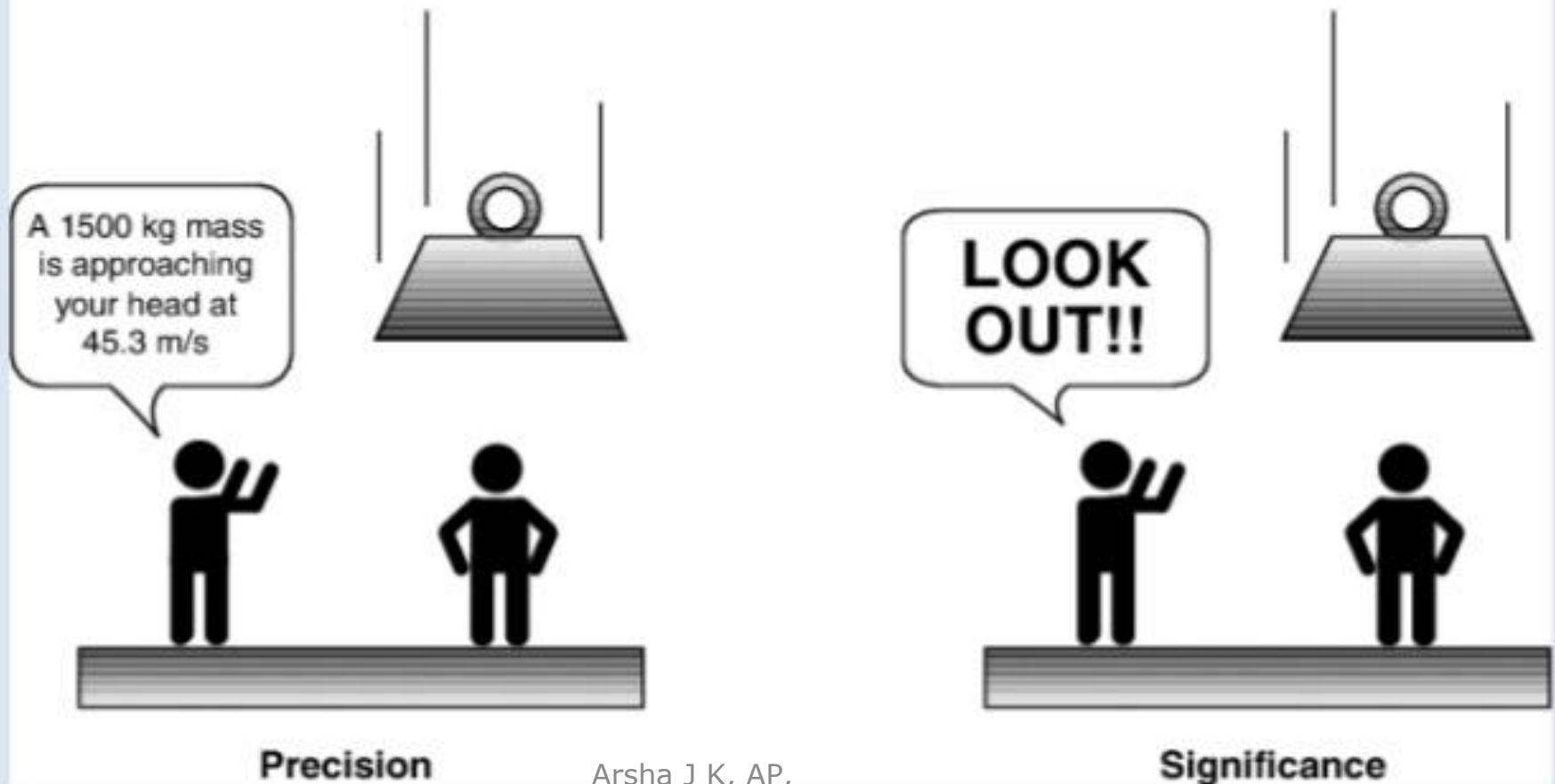
- The information revolution going on is allowing us to automate information processing tasks which require intelligence much like the industrial revolution automated manufacturing tasks
- “Soft Computing” techniques have already been applied successfully.

What is computing?

- Counting, calculating
- *The discipline of computing is the systematic study of algorithmic processes that describe and transform information: their theory, analysis, design, efficiency, implementation, and application.*
- Types of computing
 - Hard computing
 - Soft Computing

Example of Fuzzy system

Precision and Significance in the Real World



PROBLEM SOLVING TECHNIQUES

HARD COMPUTING

Precise Models

SOFT COMPUTING

Approximate Models

Symbolic
Logic
Reasoning

Traditional
Numerical
Modeling and
Search

Approximate
Reasoning

Functional
Approximation
and Randomized
Search

WHAT IS SC?

The term soft computing was proposed by the inventor of fuzzy logic, Lotfi A. Zadeh. He describes it as follows [Zadeh, 1994]:

“Soft computing is a collection of methodologies that aim to exploit the tolerance for imprecision and uncertainty to achieve tractability, robustness, and low solution cost.

Its principal constituents are fuzzy logic, neurocomputing, and probabilistic reasoning.

Soft computing is likely to play an increasingly important role in many application areas, including software engineering.

The role model for soft computing is the human mind.”

Soft Computing

- “The essence of soft computing is that unlike the traditional, hard computing, soft computing is aimed at an accommodation with the pervasive imprecision of the real world. Thus, the guiding principle of soft computing is to exploit the tolerance for imprecision, uncertainty, and partial truth to achieve tractability, robustness, low solution cost, and better rapport with reality”
- - Lotfi Zadeh



Differences between hard and soft computing

Hard Computing	Soft computing
Precisely stated analytical model	Tolerant to imprecision, uncertainty, partial truth, approximation
Based on binary logic, crisp systems, numerical analysis, crisp software	Fuzzy logic, neural nets, probabilistic reasoning.
Programs are to be written	Evolve their own programs
Two values logic	Multi valued logic
Exact input data	Ambiguous and noisy data
Strictly sequential	Parallel computations
Precise answers	Approximate answers

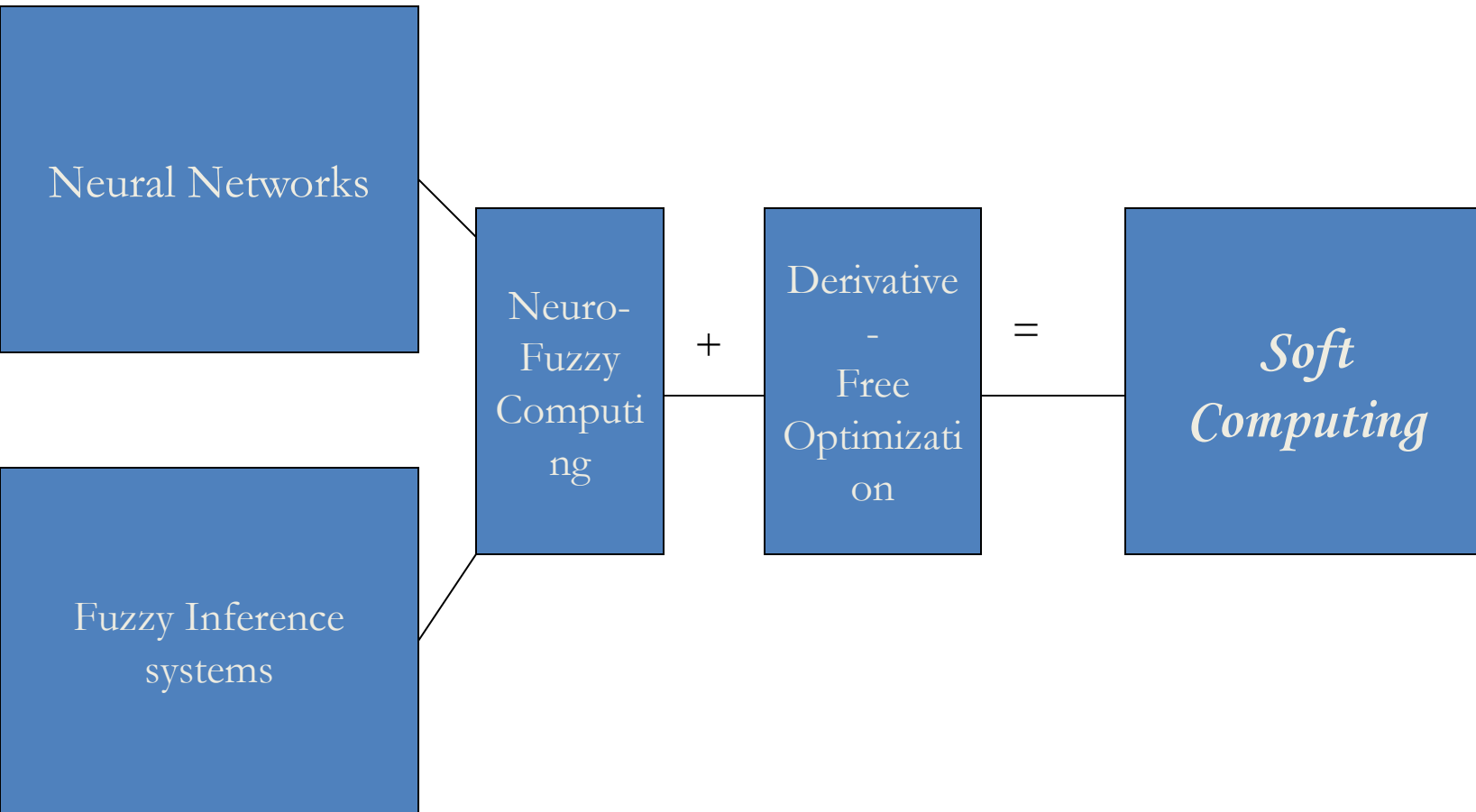
Artificial intelligence

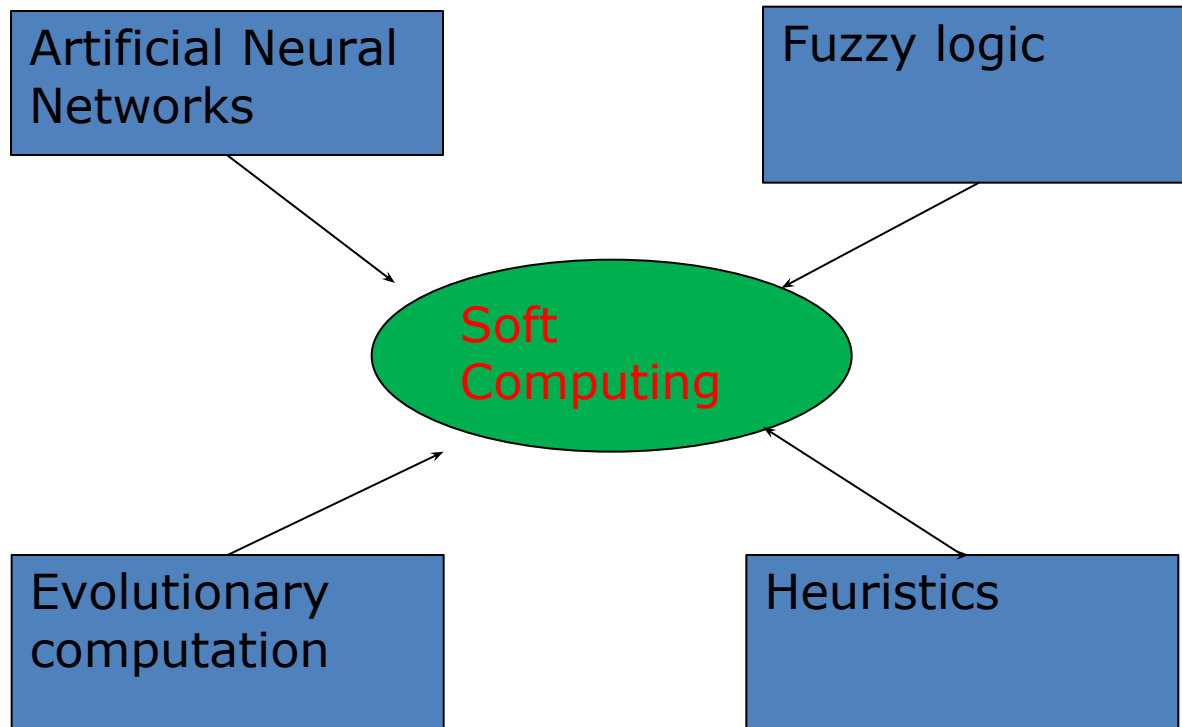
- If intelligence can be induced in machines it is called as artificial intelligence.
- Soft computing is a part of artificial intelligent techniques
- Closed related to machine intelligence / computational intelligence

OVERVIEW OF TECHNIQUES IN SOFT COMPUTING

- **Neural Networks**
- **Fuzzy Logic**
- **Genetic Algorithm**
- **Hybrid Systems**

What is Soft computing



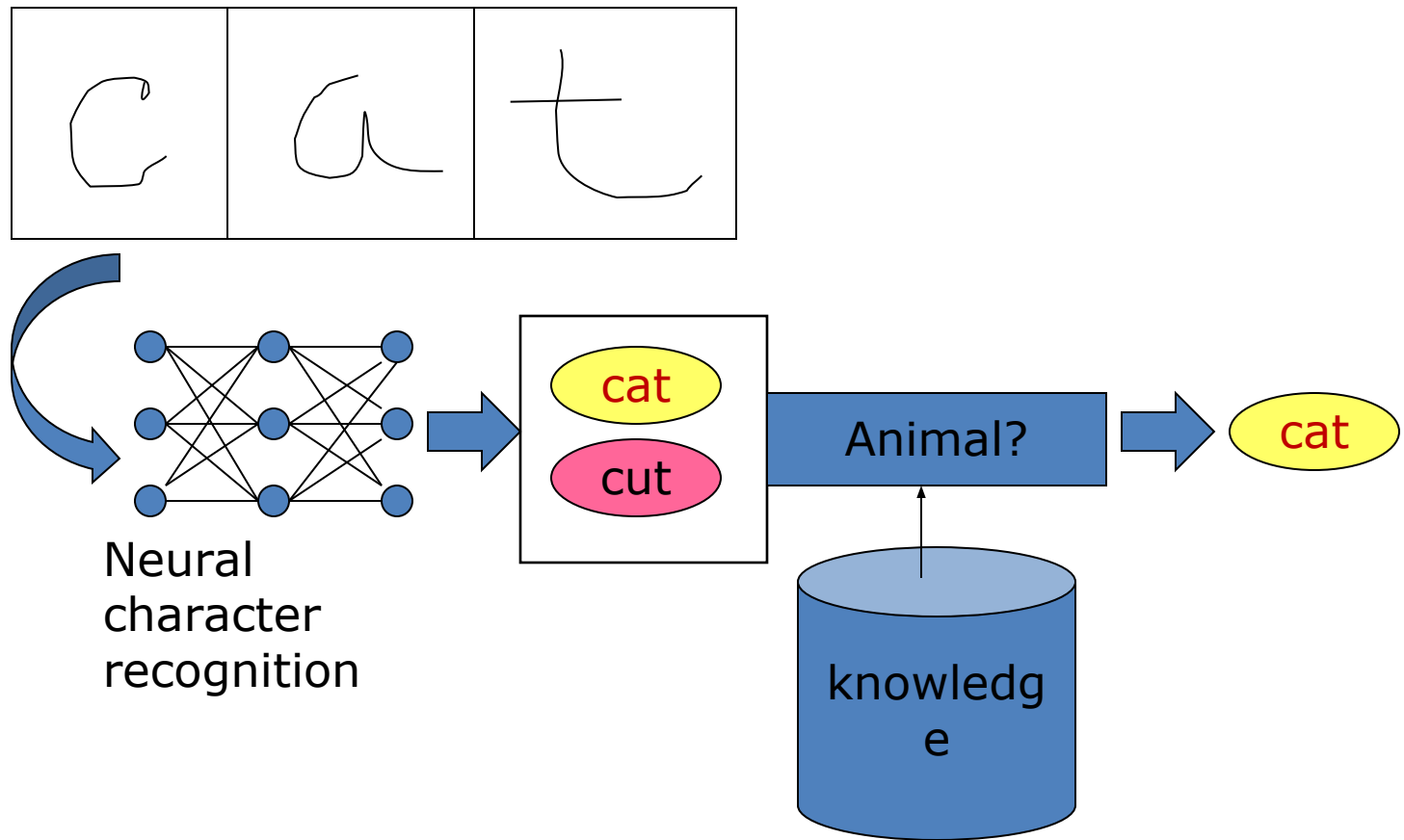


NEURAL NETWORKS

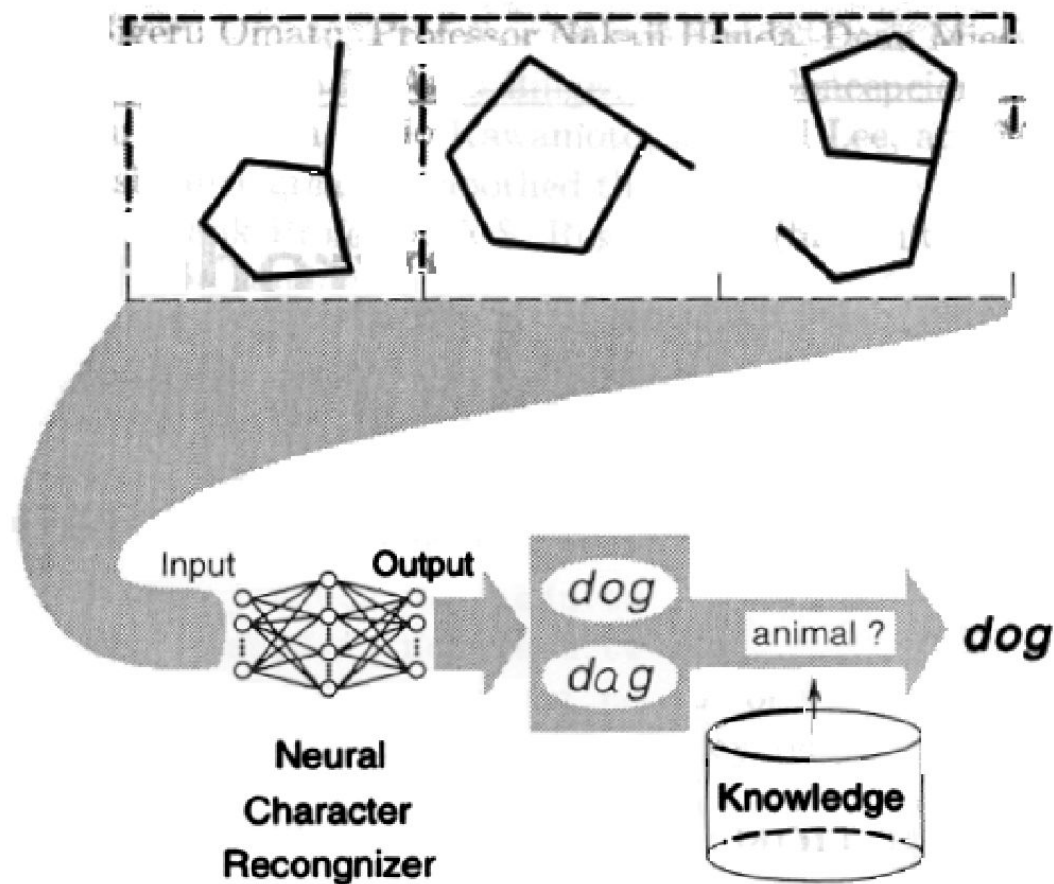
DARPA Neural Network Study (1988, AFCEA International Press, p. 60):

... a neural network is a system composed of many simple processing elements operating in parallel whose function is determined by network structure, connection strengths, and the processing performed at computing elements or nodes.

AI and Soft Computing



character recognizer



Arsha J K, AP, Computer Science and Engineering, VJCET, Vazhakulam

DEFINITIONS OF NEURAL NETWORKS

According to Haykin (1994), p. 2:

A neural network is a massively parallel distributed processor that has a natural propensity for storing experiential knowledge and making it available for use. It resembles the brain in two respects:

- Knowledge is acquired by the network through a learning process.
- Interneuron connection strengths known as synaptic weights are used to store the knowledge

According to Nigrin (1993), p. 11:

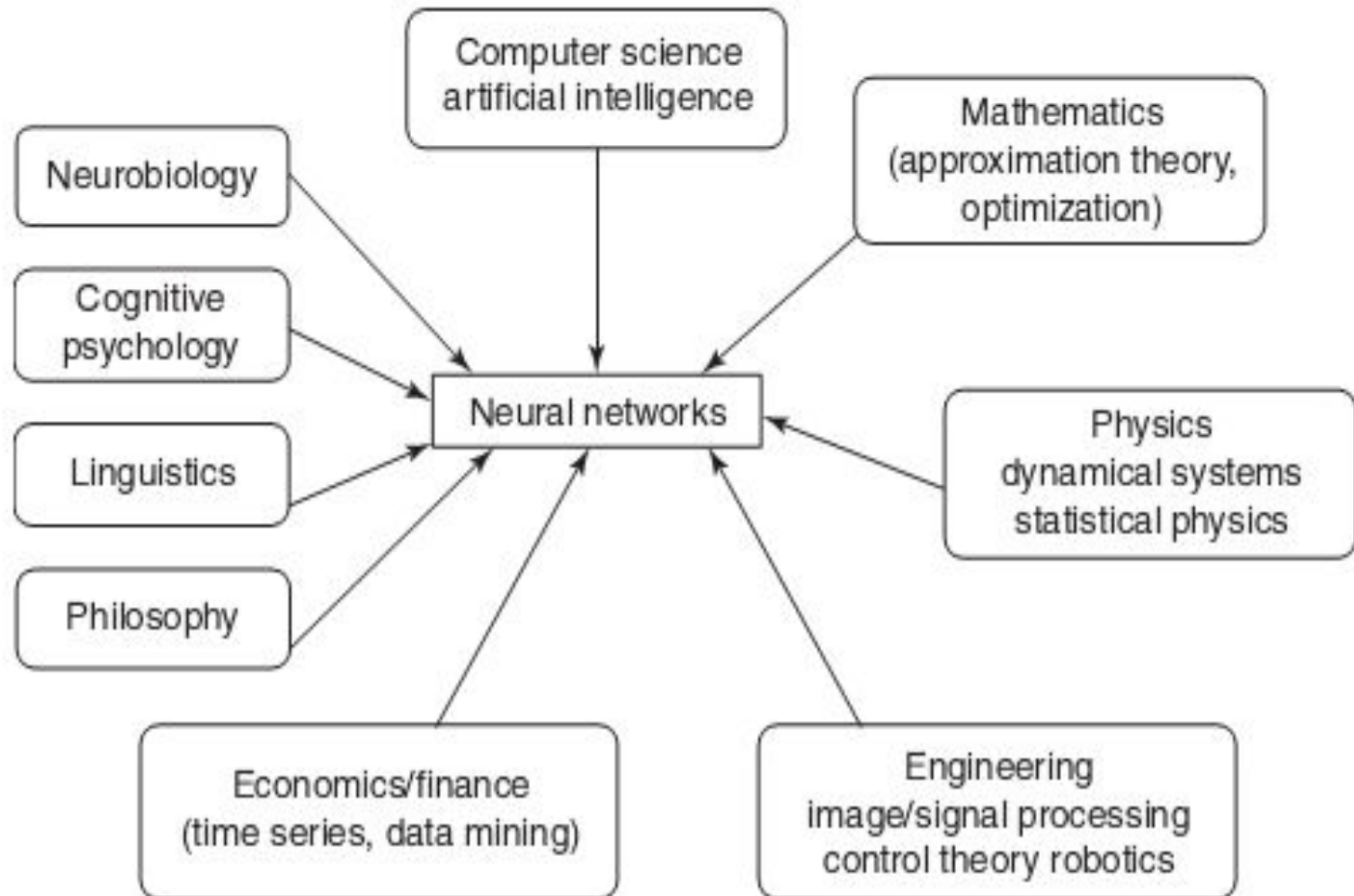
A neural network is a circuit composed of a very large number of simple processing elements that are neurally based. Each element operates only on local information.

Furthermore each element operates asynchronously; thus there is no overall system clock.

According to Zurada (1992):

Artificial neural systems, or neural networks, are physical cellular systems which can acquire, store and utilize experiential knowledge.

MULTIDISCIPLINARY VIEW OF NEURAL NETWORKS



FUZZY LOGIC

- **Origins:** Multivalued Logic for treatment of imprecision and vagueness
 - 1930s: Post, Kleene, and Lukasiewicz attempted to represent undetermined, unknown, and other possible intermediate truth-values.
 - 1937: Max Black suggested the use of a consistency profile to represent vague (ambiguous) concepts.
 - 1965: Zadeh proposed a complete theory of fuzzy sets (and its isomorphic fuzzy logic), to represent and manipulate ill-defined concepts.

FUZZY LOGIC – LINGUISTIC VARIABLES

- Fuzzy logic gives us a language (with syntax and local semantics) in which we can translate our qualitative domain knowledge.
- Linguistic variables to model dynamic systems
- These variables take linguistic values that are characterized by:
 - a label - a sentence generated from the syntax
 - a meaning - a membership function determined by a local semantic procedure

FUZZY LOGIC – REASONING METHODS

- The meaning of a linguistic variable may be interpreted as an elastic constraint on its value.
- These constraints are propagated by fuzzy inference operations, based on the generalized modus-ponens.
- An FL Controller (FLC) applies this reasoning system to a Knowledge Base (KB) containing the problem domain heuristics.
- The inference is the result of interpolating among the outputs of all relevant rules.
- The outcome is a membership distribution on the output space, which is defuzzified to produce a crisp output.

GENETIC ALGORITHM

EVOLUTIONARY PROCESS



Arsha J K, AP,
Computer Science and Engineering,
VJCET, Vazhakulam

DEFINITION OF GENETIC ALGORITHM

The genetic algorithm is a probabilistic search algorithm that iteratively transforms a set (called a population) of mathematical objects (typically fixed-length binary character strings), each with an associated fitness value, into a new population of offspring objects using the Darwinian principle of natural selection and using operations that are patterned after naturally occurring genetic operations, such as crossover (sexual recombination) and mutation.

STEPS INVOLVED IN GENETIC ALGORITHM

The genetic algorithms follow the evolution process in the nature to find the better solutions of some complicated problems. Foundations of genetic algorithms are given in Holland (1975) and Goldberg (1989) books.

Genetic algorithms consist the following steps:

- Initialization
- Selection
- Reproduction with crossover and mutation

Selection and reproduction are repeated for each generation until a solution is reached.

During this procedure a certain strings of symbols, known as chromosomes, evaluate toward better solution.

HYBRID SYSTEMS

Hybrid systems enables one to combine various soft computing paradigms and result in a best solution. The major three hybrid systems are as follows:

- Hybrid Fuzzy Logic (FL) Systems
- Hybrid Neural Network (NN) Systems
- Hybrid Evolutionary Algorithm (EA) Systems

APPLICATIONS OF SOFT COMPUTING

- Handwriting Recognition
- Image Processing and Data Compression
- Automotive Systems and Manufacturing
- Soft Computing to Architecture
- Decision-support Systems
- Soft Computing to Power Systems
- Neuro Fuzzy systems
- Fuzzy Logic Control
- Machine Learning Applications
- Speech and Vision Recognition Systems
- Process Control and So on